

Module 6: Object Oriented Programming in Dart

Enabling Assessment



Submissions

Here are your latest answers:

Answers waiting to be graded

Question 1

A _____ is an instance of a class.

Response: object

Correct answer: object

Score: 1 out of 1 Yes

Question 2

OOP is a programming _____ based on the concept of objects.

Response: paradigm

Correct answer: paradigm

Score: 1 out of 1 Yes

Question 3

The function definition variables that are designed to receive input values are called _____

Response: parameters

Correct answer: parameters

Score: 1 out of 1 Yes

Question 4

A function defined with a void return type must still explicitly include a return keyword to signal its end.

Choices:

-

True

-

False

Response: False

Correct answer: False

Score: 2 out of 2 Yes

Question 5

If a class does not define a constructor, Dart will implicitly provide a default, no-argument constructor.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 2 out of 2 Yes

Question 6

A _____ is a user-defined blueprint or template for creating objects

Response: class

Correct answer: class

Score: 1 out of 1 Yes

Question 7

The mechanism by which one class automatically acquires the properties and methods of another class is called _____.

Response: inheritance

Correct answer: inheritance

Score: 1 out of 1 Yes

Question 8

The logical flow of a program dictates that a function must be defined after the main() function, as long as it is called correctly.

Choices:

-

True

-

False

Response: False

Correct answer: False

Score: 2 out of 2 Yes

Question 9

A special constructor used to offer an alternative way to create an object, when the unnamed constructor is already defined, is called a _____ Constructor.

Response: named

Correct answer: Named

Score: 1 out of 1 Yes

Question 10

A function provided automatically by Dart, such as toString() or print(), is referred to as a _____ function.

Response: built-in

Correct answer: built-in

Score: 1 out of 1 Yes

Question 11

Polymorphism is the OOP concept that describes how one class automatically inherits properties and methods from another.

Choices:

-

True

-

False

Response: False

Correct answer: False

Score: 2 out of 2 Yes

Question 12

The process of creating an object from a class is called _____ the class.

Response: instantiating

Correct answer: instantiating

Score: 1 out of 1 Yes

Question 13

Question: Classes and Objects are the cornerstones of the Object-Oriented Programming (OOP) paradigm in Dart. Discuss the distinct conceptual roles of a Class and an Object, and explain how their relationship facilitates the development of robust, reusable, and organized software.

Response: A class in Dart serves as a blueprint or template that defines the structure, properties, and behaviors (including methods) for a type of entity, while an object is a concrete instance of that class, representing a specific realization with its own unique state and data. Their relationship allows developers to create multiple objects from a single class, enabling the reuse of common code structures across instances without duplication. This synergy promotes robust software through encapsulation and error isolation, reusability via modular design, and organization by modeling real-world entities hierarchically, making code more maintainable and scalable.

Score: not graded yet

Question 14

The ability of an object to take on many forms is known as _____

Response: polymorphism

Correct answer: polymorphism

Score: 1 out of 1 Yes

Question 15

Arguments are the variables listed in a function's definition, while parameters are the specific values passed during the call.

Choices:

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True

-

False

Response: False

Correct answer: False

Score: 2 out of 2 Yes

Question 16

The purpose of a Constructor is to create and initialize a new instance of a class.

Choices:

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True

-

False

Response: True

Correct answer: True

Score: 2 out of 2 Yes

Question 17

Functions are used to promote code reusability, organization, and ease of debugging.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 2 out of 2 Yes

Question 18

Instance variables (data) and _____ (code) are the two components typically contained within an object.

Response: methods

Correct answer: methods

Score: 1 out of 1 Yes

Question 19

Encapsulation is the OOP principle that restricts direct access to a class's internal data, forcing interaction through defined methods.

Choices:

-

True

-

False

Response: True

Correct answer: True

Score: 2 out of 2 Yes

Question 20

In Dart's OOP, an Object is the fundamental blueprint or template used to define fields and methods.

Choices:

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True

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False

Response: False

Correct answer: False

Score: 2 out of 2 Yes

Question 21

A class is restricted to having only one constructor, which must be the default, no-argument version.

Choices:

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True

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False

Response: False

Correct answer: False

Score: 2 out of 2 Yes